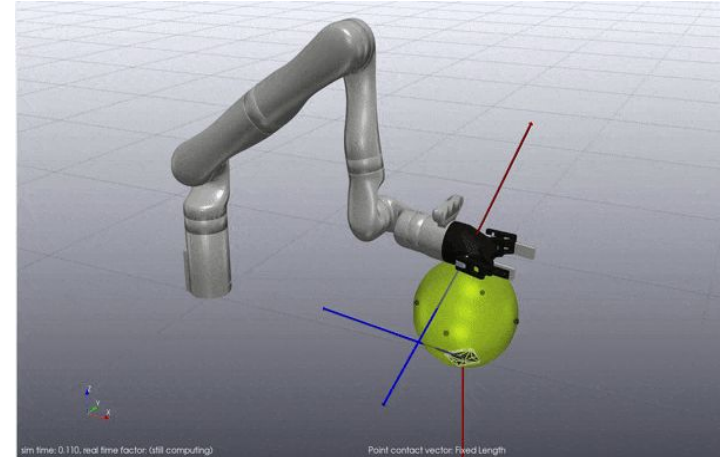


Improving Diffusion Policies with Latent Noise Steering for Planar Manipulation

Gabriel Olin, Benji Li

Non-Prehensile Manipulation

- Manipulation without grasping (no stable enclosure of the object)
- Many real-world tasks involve objects that are too **large, heavy, slippery**, or **poorly shaped** to grasp reliably.
- Robot influences object motion through transient contact, forces, friction, and environmental interactions.
 - Pushing
 - Sliding
 - Rolling

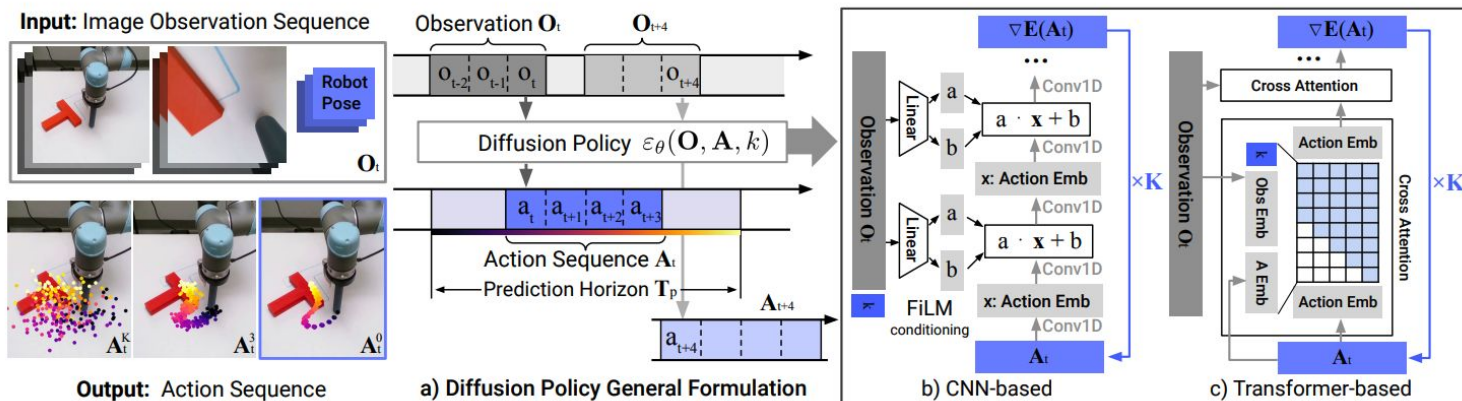


Denoising Diffusion Probabilistic Models

- Learns a **generative model** by **gradually adding noise to data** and then training a neural network to **reverse the noising process**, recovering clean samples step-by-step.

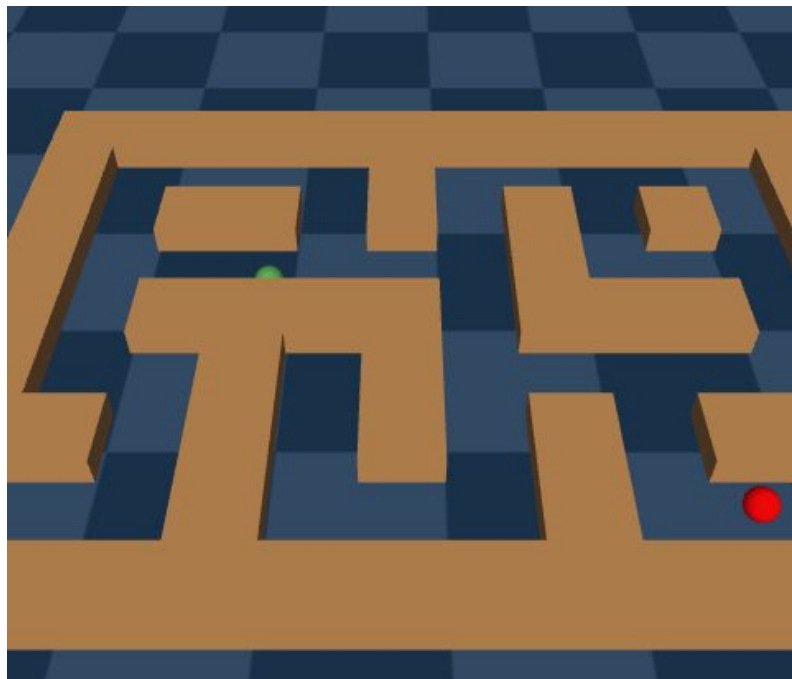
Useful for behavior cloning based robot manipulation policies

- Captures multi-modal action/trajectory distributions for contact-rich tasks
- Produces smooth, stable action sequences; robust for long-horizon imitation and planning
- Learns a conditional posterior over trajectories given observations, implicitly capturing task dynamics without requiring explicit or accurate contact models (unlike MPC)



Navigation Environment (PointMaze_Large-v3)

- Validated diffusion policy on common offline RL datasets (D4RL)
- Point Maze environment requires long horizon reasoning
 - Navigate force controlled ball through maze
 - Trained on Q-learning + PD control expert



Environment

- Push-T SE(2) task:
 - Use point-contacts from the rod to push and reorient the T object to the goal position
 - Success when key points on the T lie within a predefined L2 distance of the goal position (Fig 2)
- Observation Space:
 - (end_eff position, T pose, goal T pose)
 - Absolute (x,y,z) position, quaternion orientation
- Action Space:
 - $\Delta(x,y,z)$ position, joint angles computed via IK and tracked by a PD controller
 - Position actions empirically work more effectively than velocity actions for diffusion models (Chi et al, 2023)
- Reward
 - Negative sum of position and orientation error

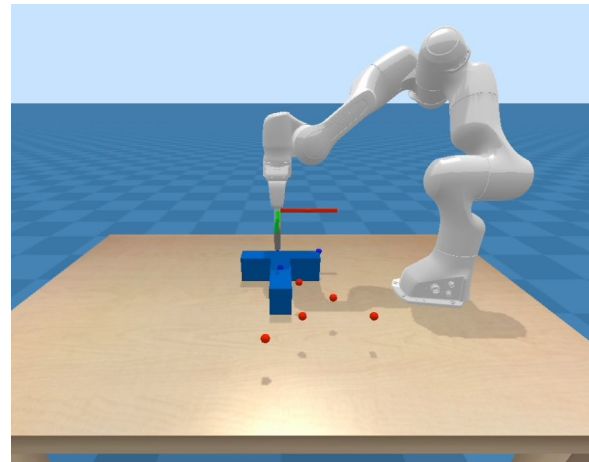


Fig 1: Pose at start of demonstration

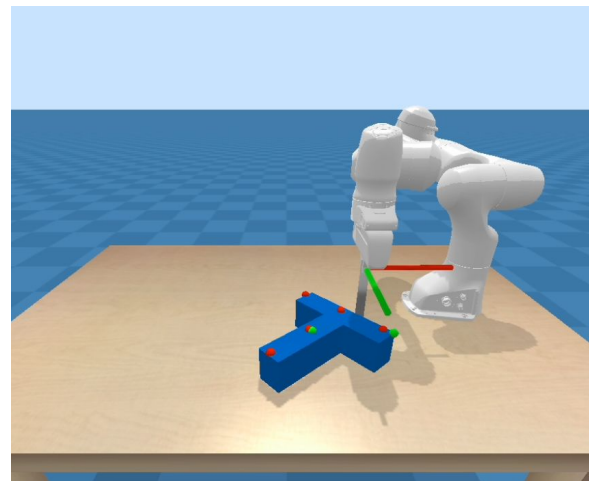
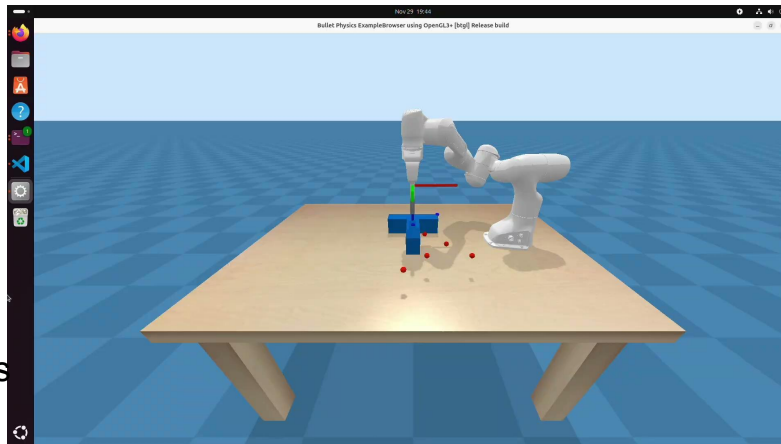


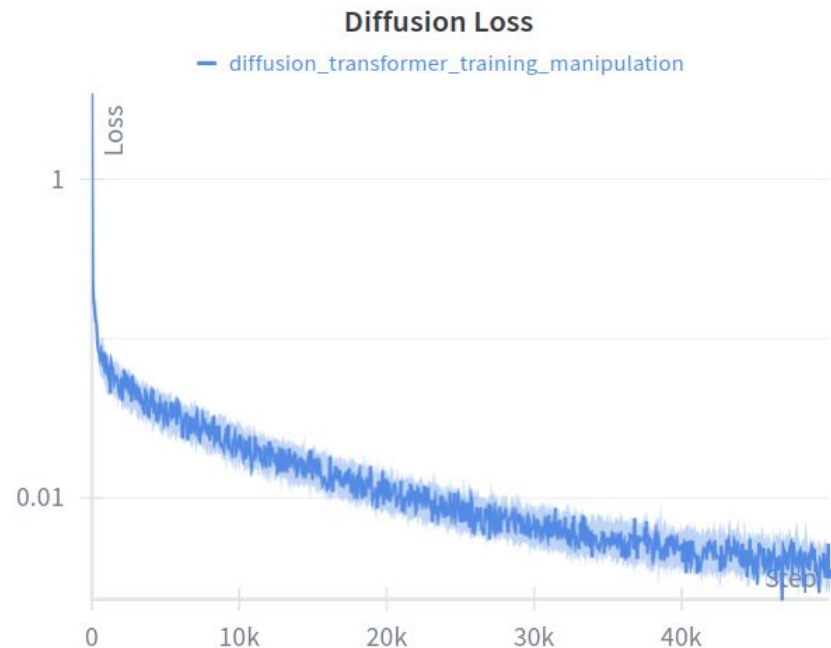
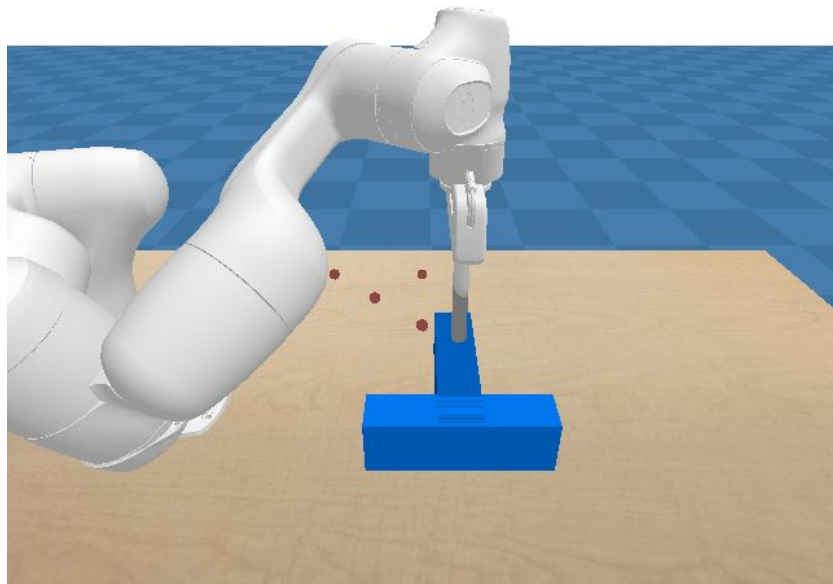
Fig 2: Finished demonstration

Data Collection

- Teleoperation via joystick controller, 100 demonstrations collected per dataset
- Dataset 1:
 - Predefined start and end poses for the end-effector and T
- Dataset 2:
 - Randomized start and end poses for end-effector and T (within a predefined workspace)



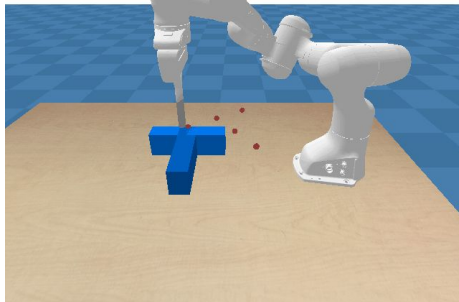
Results - Single Goal Policy



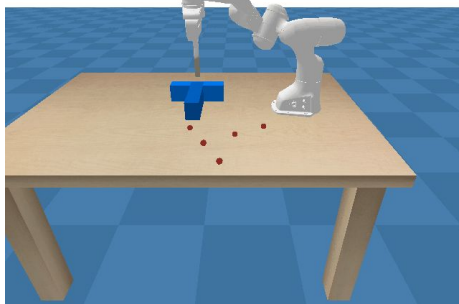
$$\mathcal{L}(\theta) = \mathbb{E}_{a_0, s, t, \epsilon \sim \mathcal{N}(0, I)} \left[\|\epsilon - \epsilon_\theta(a_t, s, t)\|_2^2 \right]$$

Results - Multi Goal Policy

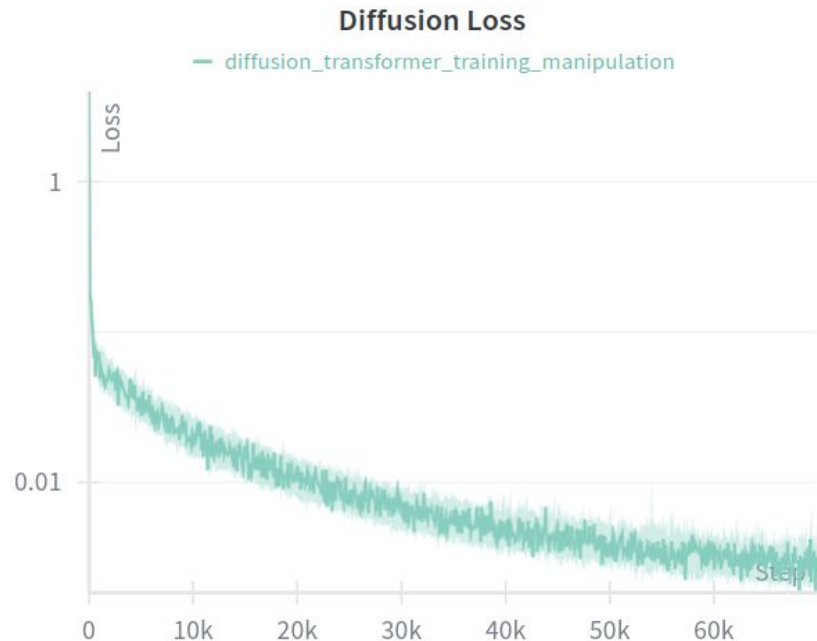
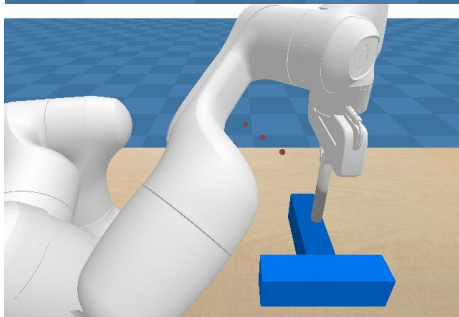
Goal 1:
Similar Position/Orientation



Goal 2:
180 Degree Spin



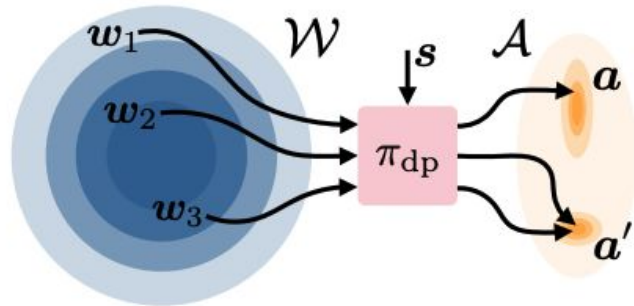
Goal 3:
Edge of Table



$$\mathcal{L}(\theta) = \mathbb{E}_{a_0, s, t, \epsilon \sim \mathcal{N}(0, I)} \left[\|\epsilon - \epsilon_{\theta}(a_t, s, t)\|_2^2 \right]$$

Diffusion Steering via Reinforcement Learning

- Vanilla diffusion samples initial noise from a constant Gaussian distribution
- Can we learn to steer the diffusion policy toward desired goals by changing the parameters of the initial noise Gaussian?

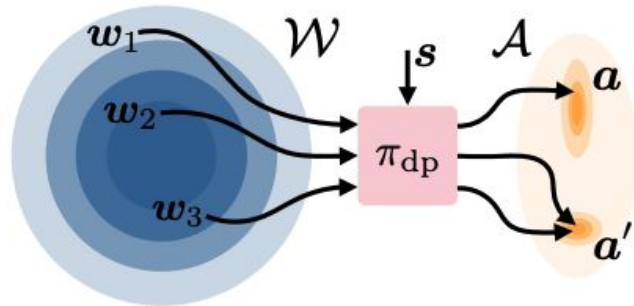


Algorithm 1 Noise-Aliased Diffusion Steering via Reinforcement Learning (DSRL-NA)

- 1: **input:** pretrained diffusion policy π_{dp}^w , offline data $\mathcal{D}_{off}$ and/or online environment $\mathcal{M}$
 - 2: Initialize replay buffer $\mathfrak{B} \leftarrow \mathcal{D}_{off}$, $\mathcal{A}$ -critic Q^A , latent-noise critic Q^w , latent-noise actor π^w
 - 3: **for** $t = 1, \dots, T$ **do**
 - 4: Update Q^A : $\min_{Q^A} \mathbb{E}_{(s, a, r, s') \sim \mathfrak{B}, a' \sim \pi_{dp}^w(s', \pi^w(s'))} [(Q^A(s, a) - r - \gamma \bar{Q}^A(s', a'))^2]$
 - 5: Update Q^w : $\min_{Q^w} \mathbb{E}_{s \sim \mathfrak{B}, w \sim \mathcal{N}(0, I)} [(Q^w(s, w) - Q^A(s, \pi_{dp}^w(s, w)))^2]$
 - 6: Update π^w : $\max_{\pi^w} \mathbb{E}_{s \sim \mathfrak{B}} [Q^w(s, \pi^w(s))]$
 - 7: **if** access to online environment $\mathcal{M}$ **then**
 - 8: Sample latent-noise action $w_t \sim \pi^w(s_t)$ and compute $a_t \leftarrow \pi_{dp}^w(s_t, w_t)$
 - 9: Play a_t in $\mathcal{M}$, observe r_t and next state s_{t+1} , and add (s_t, a_t, r_t, s_{t+1}) to $\mathfrak{B}$
-

Diffusion Steering via Reinforcement Learning

- Vanilla diffusion samples initial noise from a constant Gaussian distribution
- Can we learn to steer the diffusion policy toward desired goals by changing the parameters of the initial noise Gaussian?



Benefits:

- Assume only black box access to pretrained diffusion policy
- Enables efficient online adaptation of policy without collecting more human demos

Diffusion Steering via Reinforcement Learning

Transforms original MDP to latent action MDP:

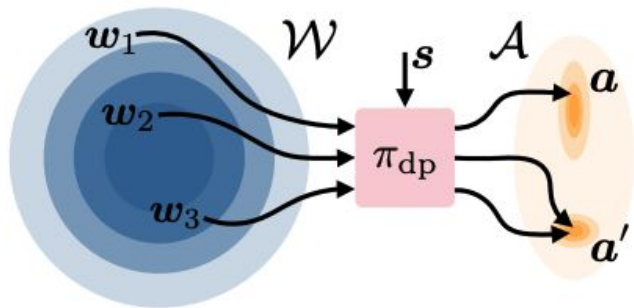
$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, p_0, r, \gamma) \longrightarrow \mathcal{M}^{\mathcal{W}} := (\mathcal{S}, \mathcal{W}, P^{\mathcal{W}}, p_0, r^{\mathcal{W}}, \gamma)$$

$\pi^{\mathcal{W}}(\mathbf{s})$ Actor $\rightarrow$ Outputs initial noise

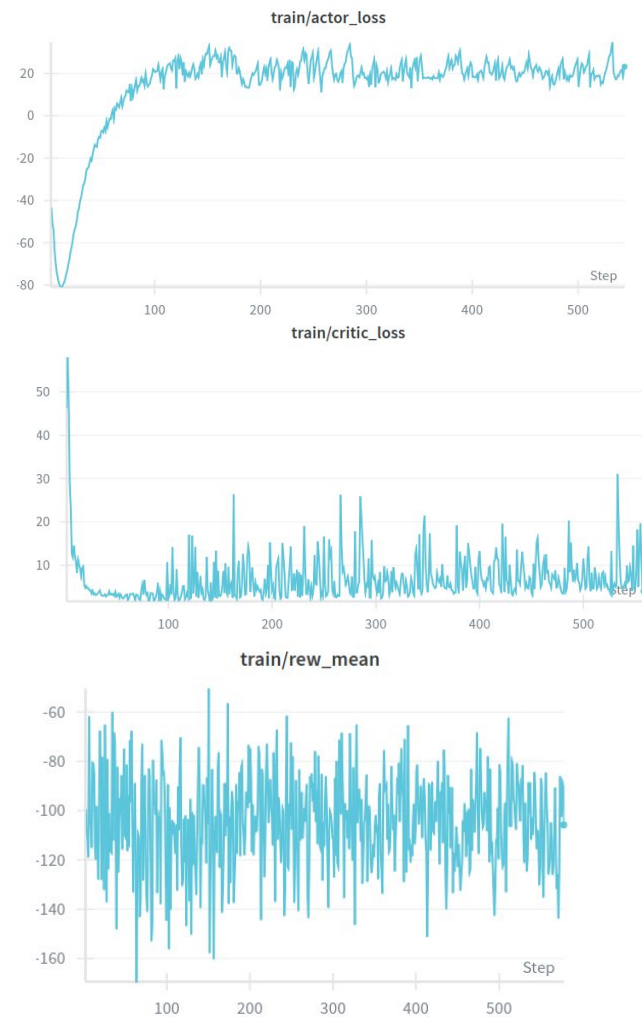
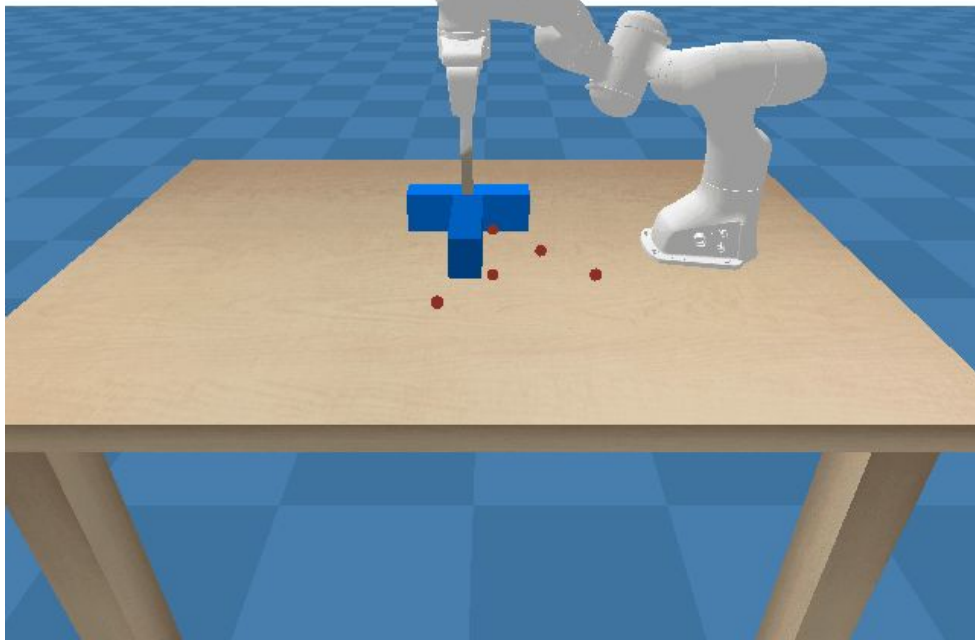
$\pi_{\text{dp}}^{\mathcal{W}}(\mathbf{s}, \pi^{\mathcal{W}}(\mathbf{s}))$ Action sampled from diffusion policy with optimized noise

$Q^{\mathcal{A}}$ Critic in original MDP

$Q^{\mathcal{W}}$ Critic in latent action MDP



Results - DSRL Tuned Policy



Future Work / Limitations

Limitations

- Policy struggles to recover from OOD states
- Doesn't generalize well to OOD start and goal locations
- Small dataset for multi-goal setting inhibits effectiveness of learned policies

Future Work

- Combine with Goal Conditioned RL techniques (HER, AM) to improve performance on long horizon tasks
- Incorporate search or expert guidance to help recover from OOD states and improve generalizability

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